INCH POUND

MIL-STD-869E
10 February 2022

SUPERCEDING
MIL-STD-869D
01 March 2007

DEPARTMENT OF DEFENSE
STANDARD PRACTICE

THERMAL SPRAY

This document is reactivated after 10 February 2022
and may be used for new and existing designs and acquisitions.

AMSC: NA

AREA: MFFP

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard provides guidance on the process of thermal spraying for the Air Force repair, acquisition, and manufacture of parts and spare parts on all military aircraft.

3. Comments, suggestions, or questions on this document should be addressed to the DAF Landing Gear Program, Attn: Engineering at AFMC 417 SCMS/GUEA, Hill AFB, UT 84056-5228 or emailed to 417SCMS.GUEA.workflow@us.af.mil. All technical questions for this document contact DAF 309th CMXG 5760 Southgate Ave. Bldg. 511 Hill AFB UT 84056-5228 or emailed to 309.CMXG.Workflow@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
	SUMMARY OF VERSION E MODIFICATIONS	vi
1	SCOPE	1
1.1	Scope	1
	Process	1
	Classification	1
	Types	
2	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government documents	1
2.2.1	Specifications, standards, and handbooks	1
2.2.2	Other Government documents, drawings, and publications	2
2.3	Non-Government publications	3
2.4	Order of Precedence	3
3	DEFINITIONS	4
4	GENERAL REQUIREMENTS	4
4.1	Facility requirements	4
4.1.1	Environmental controls	4
4.1.2	Booth controls	4
4.2	Equipment requirements	4
4.2.1	Control system	4
4.2.2	Powder feeder	5
4.2.3	Multi-axis industrial robot and controller	5
4.2.4	Supply of oxygen	5
4.2.5	Supply of fuel	5
4.2.6	Dust extractor and bag-house filtration system	5
4.2.7	Dry, oil-free compressed air	5
4.2.8	Water cooling	5
4.2.9	Infrared pyrometer	5
4.2.10	Part holding and turning device	5
4.2.11	Wire flame spray system	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

CONTENTS (continued)

PARAGRAPH		PAGE
4.2.12	Powder flame spray system	6
4.2.13	Plasma spray system - Powder	6
4.2.14	Electric wire (twin wire arc) spray system	6
4.2.15	Abrasive blasting equipment	7
4.3	Material requirements	7
4.3.1	Coating material	7
4.3.2	Bond strength	7
4.3.3	Thermal spray wires	7
4.3.4	Thermal spray powders	7
4.3.5	Gases	7
5.	DETAILED REQUIREMENTS	8
5.1	Thermal spray repair operations - Procedure	8
5.1.1	Part cleaning	8
5.1.2	Masking	8
5.1.3	Abrasive grit blasting	9
5.1.4	Preheating	9
5.1.5	Cooling while spraying	10
5.1.6	Spraying parameters	10
5.1.7	Spraying	10
5.1.8	Visual inspection	10
5.1.9	Removal of masking - Cleaning	11
5.1.10	Sealing sprayed coatings	11
5.1.11	Finish machining and grinding	11
5.1.12	Stripping	11
5.2	Quality control	11
5.2.1	Record Keeping	11
5.2.2	Coating thickness	11
5.2.3	Workmanship	11
5.2.4	Testing	12
5.2.4.1	Bend strength	12
5.2.4.2	Bond strength	13
5.2.4.3	Metallography	13
5.3	Quality	14
5.3.1	Pre-production tests	14

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

CONTENTS (continued)

PARAGRAPH			PAGE
5.3.2		Acceptance test frequency	14
5.3.3		Acceptance test completion	14
6	NOTES		15
6.1		Intended use	15
6.2		Subject term (key word) listing	15
6.3		Changes from previous issue	15
FIGURES			
1		Bend test specimen.	12
2		Acceptance standard for bend test specimens.	13
TABLES			
I		Types of masking tapes.	9
APPENDIX			
A		Sample plasma coating process operations sheet.	16
A.1	SCOPE		16
CONCLUDING MATERIALS			17

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

SUMMARY OF VERSION E MODIFICATIONS

1. Changed the Uniform Resource Locator (URL) for the Acquisition Streamlining and Standardization Information System (ASSIST) database throughout the document.

2. Updated foreword: added updated contact information.

3. Updated concluding material: added updated custodian and review activities.

4. The following modifications to MIL-STD-869E have been made:

PARAGRAPH		MODIFICATIONS	
	Forward		Updated
	Change summary page		Added
	1.1		Changed
	1.3.1		Changed
	1.3.1.1		Changed
	1.3.1.2		Changed
	1.3.1.3		Changed
	1.3.1.4		Changed
	2.1		Changed
	2.2.1		Changed
	2.2.2		Added
	2.3		Changed
	2.4		Added
	3.		Changed
	3.1 – 3.8		Added
	4.1		Changed
	4.1.1		Changed
	4.1.2		Changed
	4.1.3		Changed
	4.1.3.a. – e.		Changed
	4.1.4. and 4.1.4.a. – e.		Changed
	4.1.5 – 4.1.8		Changed
	4.2		Changed
	4.2.1 – 4.2.15		Added
	4.3		Changed
	4.3.1 – 4.3.5.1		Added
	4.4 – 4.7.b.		Changed
	5.1 – 5.1.8		Changed
	5.1.9 – 5.1.12		Added
	5.2.1 – 5.2.4		Changed
	5.2.3 – 5.3.3		Added
	6.1		Changed
	Figure 1		Changed
	Figure 2		Added
	Table I		Added
	Appendix A		Added
	CONCLUDING MATERIAL		Changed

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard covers provides guidance on the material, procedures, processes, and equipment for the thermal spraying of aerospace components.

1.2 Process. Thermal spraying is defined as the process of melting materials in a heating zone to a molten or semi-molten state and propelling them onto a substrate to form a coating. This process is approved for building up worn areas on parts, and for abrasion resistance and/or corrosion protection. It is not intended to rebuild damage by nicks, gouges, etc., without building an entire area.

1.3 Classification.

1.3.1 Types. There are four (4) types of thermal spraying covered in this standard.

1.3.1.1 Type I. *Wire flame spray:* Spraying materials in a wire form using oxygen-acetylene gas, hydrogen gas, propane, etc.

1.3.1.2 Type II. *Powder flame spray:* Spraying materials in a powder form, using oxygen, acetylene, or hydrogen gas.

1.3.1.3 Type III. *Plasma spray:* Spraying materials in a wire or powder form, utilizing a non-transferable electric arc with nitrogen or argon, with helium and/or hydrogen as a secondary gas with or without hydrogen.

1.3.1.4 Type IV. *Electric arc (twin wire arc) spray:* Arc spraying system, simultaneously and continuously feeds two (2) metalizing wires at a uniform rate of speed through electrically charged contact blocks, propelled using compressed gas (air).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited within this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. If the solicitation or contract does not specify an issue or revision, follow the current version of the applicable document(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

FEDERAL STANDARDS

BB-A-106	Acetylene, Technical, Dissolved
BB-H-886	Hydrogen

(Copies of these documents are available online at https://assist.dla.mil)

COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-59155	Nitrogen, High purity, Special Purpose
A-A-59503	Nitrogen, Technical

(Copies of these documents are available online at https://assist.dla.mil)

DEPARTMENT OF DEFENSE STANDARDS

MIL-DTL-18455	Argon, Technical
MIL-DTL-5002	Surface Treatments and Inorganic Coatings For Metal Surfaces of Weapons Systems
MIL-STD-866	Grinding of Chrome Plated Steel and Steel Parts Heat Treated to 180,000 PSI or Over
MIL-STD-1504	Abrasive Blasting

(Copies of these documents are available online at https://assist.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODEOF FEDERAL REGULATIONS (CFR)

29 CFR 1910	Occupational Safety and Health Standards (OSHA)

(Copies of this document are available online at https://www.osha.gov/.

2.3 Non-Government publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM C 633	Standard Test Method for Adhesion or Cohesion Strength of Thermal Spray Coatings
ASTM E 1920	Standard Guide for Metallographic Preparation of Thermal Sprayed Coatings
ASTM E 2109	Standard Test Methods for Determining Area Percentage Porosity in Thermal Sprayed Coatings

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

ASTM E 384	Standard Test Method for Microindentation Hardness of Materials

(Copies of these documents are available online at www.astm.org.)

PRAXAIR GASES

Praxair B-72	Propane, Pure C8H8

(Copies of this document are available on-line at https://www.praxaireusa.com/.)

SOLVAY

FM© 1000	Film adhesive, with and without primer

(Copies of this document are available on-line at https://www.solvay.com)

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-4	Oxygen

(Copies of this document are available online at https://portal.cganet.com/Publication/index.aspx)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 6454	Steel, Sheet, Strip, and Plate 1.8Ni – 0.80Cr – 0.25Mo (0.38 – 0.43C)
AMS 6351	Steel, Sheet, Strip, and Plate 0.95Cr – 0.20Mo (0.28 – 0.33C)

(Copies of these documents are available at https://www.sae.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this standard takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

3 DEFINITIONS

3.1 100% Duty cycle. The capability for a power supply to run at 100% capacity throughout the spraying process.

3.2 Applicable documents. Specifications, Standards, and Handbooks, Federal Specifications, Non-Government documents, Technical Orders (TO), Work Control Documents (WCD), Air Force Engineering Drawings.

3.3 Bond coat. A layer of material applied to help assure bonding of coating material to the part, similar in intent to paint primers

3.4 Engineering Source Authority (ESA). The requiring Activity's cognizant engineering authority responsible for making the engineering decisions for the requirements.

3.5 High strength. High Strength materials have been heat treated to 180 ksi (1.24×10^9 Pa) tensile strength or greater.

3.6 Manufacturer's recommendations. Instruction Manuals, Bulletins, and Technical Data provided by the manufacturer, regarding equipment and materials.

3.7 Occupational Safety and Health Administration (OSHA). 29 CFR §1910 *OSH Standards*. Responsible for establishing and enforcing safety and health standards in the workplace.

3.8 Working distance. Working Distance refers to the distance between the gun and the substrate to be sprayed or "gun-to-work" distance, and determined by the manufacturer's recommendations for the coating material being sprayed.

4 GENERAL REQUIREMENTS

4.1 Facility requirements

4.1.1 Environmental controls. The thermal spray facility shall be temperature and humidity controlled, and free from leaks. These factors depend highly on material and processes used and the impact on coating quality and acceptability.

4.1.2 Booth environment. A soundproof booth shall be capable of maintaining a non-condensing atmosphere in accordance with OSHA or equivalent agency guidelines. Follow material manufacturer recommendations for humidity and other atmospheric conditions.

4.2 Equipment requirements. The following parameters shall be adhered to using the following parameters to produce quality coatings:

4.2.1 Control system. The system shall be computer controlled, when feasible, and when used shall be fitted with an automated device for regulating the gas(es) and fuel(s) used to operate the torch.

4.2.1.1 Indicating devices. When indicating devices are used as system controls applicable to the process gases and fuels, the device shall have accuracy, at a minimum, as follows:

Pressure gauges	± 4.0% of full scale
Flow meters	± 4.0% of full scale

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

4.2.2 Powder feeder. The powder feeder unit shall be capable of supplying a metered and consistent flow of powdered material to the gun.

4.2.3 Multi-axis industrial robot and controller. An industrial robot shall be used, whenever possible or practical, to manipulate the gun to ensure even spraying. When a robot is used, it shall be securely mounted to ensure stability during operation and an additional axis used, as required, with the robot controlled system for part manipulation or other needs. Mount the torch on an automated manipulating device during the deposition process to maintain a constant working distance and traverse rate, or maintained in a fixed position with the component mounted on a manipulating device that will maintain a constant working distance. Integrated process recording is recommended.

4.2.4 Supply of oxygen. Oxygen shall be supplied in bulk. When in liquid form, it shall be vaporized to gas and shall be sufficient in volume and pressure to achieve required operating conditions.

4.2.5 Supply of fuel. Fuel shall be supplied in bulk, and supplied to meet pressure and flow requirements.

4.2.6 Dust extractor and bag-house filtration system. Local guidelines shall be followed for a filtration system for dust removal. The system is commonly located on the outside the building. The air is blown or pulled into the filtration system to help extract the air from the booth laden with overspray or particles that have failed to stick to the surface.

4.2.7 Dry, oil-free compressed air. Air cooling is one method used to prevent coated components from overheating and equipment controls. Compressed air shall be clean, dry, and oil free. A compressed air system shall be maintained and shall include an integrated refrigerated dryer rated for 35 to 39 °F (1.7 to 3.9 ºC) dew point, 3.9 x 10-5 inch (1 micron) particulate filtration, and 0.1 ppm oil removal.

4.2.8 Water cooling. Various thermal spray systems require chilled conditioned water for torch tooling. The heat exchanger shall be sufficient to control the water temperature to within Original Equipment Manufacturer (OEM) recommendations.

4.2.9 Infrared pyrometer. Temperatures shall be monitored during the application of thermal spray coatings. Non-contact temperature measurement equipment with a minimum resolution of 1 °F (-17.2 °C) and response time of less than one (1) second shall be utilized. This measurement shall be verified by calibrated temperature probes.

4.2.9.1 Part set-up and prototyping. Temperature measurement set-up and values shall be verified during part set-up and prototyping for the no-touch temperature sensor with a calibrated contact probe type thermocouple. The thermocouple shall be held firmly against the heated part and adjusted so the no-touch sensor equipment temperature readings are within 5% of the touch probe reading. Adjust the no touch system as needed for each part or surface.

4.2.10 Part holding and turning device. The application of thermal spray coatings onto aerospace components requires the use of equipment to rotate parts at controlled speeds during the spray process. The position is horizontal, vertical, or both. The rotating equipment shall be capable of maintaining constant set speeds, regardless of load, and shall provide RPM feedback data to the system controller.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

4.2.11 Wire flame spray system. The wire system shall include an electronically controlled gun. Gas and oxygen regulators shall be two (2) stage regulators. Fuel gas flow meters and regulators shall be equipped with flash back valves. Air pressure regulators shall be an OSHA approved type. Gas hoses, wire controller, straightening units, and engineering approved optional equipment shall complete the system.

4.2.12 Powder flame spray system. The powder flame spray system shall include a spray gun capable of delivering and maintaining a constant flow rate. Regulators shall be the two (2) stage type. Fuel gas flow meters and regulators shall be equipped with flash back valves. Air regulators shall be OSHA approved. Gas hoses and engineering approved optional equipment shall complete the system.

4.2.13 Plasma spray system – Powder. The plasma spray system shall include a dispensing gun capable of operating on both a primary and secondary gas, with powder feed and controlled electric arc.

a. The plasma gun shall be capable of spraying coatings of powdered materials that have been heated to a plastic or molten state by a highly ionized gas stream passing through a non-transferable electric arc capable of reaching elevated temperatures.

b. The direct current (DC) power supply unit shall be sufficient for current at 100% duty cycle.

c. As plasma equipment settings are developed for new recipes, they shall be recorded so set-up time to duplicate a particular coating on a particular part is minimized. Appendix A shows a sample *Plasma Coating Process Operation Sheet*. These records shall serve as a guide when setting up equipment.

4.2.14 Electric wire (twin wire arc) spray system. The arc spray system shall consist of a gun capable of simultaneously feeding two (2) wires at a controlled rate of speed to deliver a fine atomized spray onto a prepared surface.

a. The gun shall be hand held or machine mounted and be capable of controlling wire feed rate, and shall also be capable of operating solely on compressed air for dust offs.

b. The DC power supply unit shall be capable of delivering amperage and voltage at 100% duty cycle.

c. The control unit shall be compatible with automatic traverse equipment, and include air, amperage, and voltage controls, and monitors.

d. Wire control units shall include wire straighteners.

e. Air hoses, wire lines, power cables, and engineering approved optional equipment shall complete the system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

4.2.15 Abrasive blasting equipment. Abrasive blasting equipment shall be used to remove oxidation and roughen areas to be sprayed IAW MIL-STD-1504. Parts shall be previously cleaned in accordance with MIL-DTL-5002. Particular care shall be taken to keep abrasive grit from contaminating the surrounding area. Unless otherwise specified by the ESA, blasting equipment shall be used solely for preparing parts for spraying, and the manufacturer's recommended procedures shall be followed.

4.3 Material requirements.

4.3.1 Coating material. The ESA will specify the coating material. Metallurgical laboratory testing shall be performed, as required.

4.3.2 Bond strength. Acceptable thermal spray coatings often have bond strengths ranging from 3.0 to 12.0 ksi ($2.1 \times 10^7 - 8.3 \times 10^7$ Pa). Coatings shall only be used where the bond strength of the coating has been approved for a specific application by the ESA.

4.3.3 Thermal spray wires. Unless otherwise specified by the ESA, only wires specifically manufactured for thermal spray shall be used. The chemical composition of the material to be used shall be specified in the contract or the material purchase order, and can be found in the manufacturer's recommendations.

4.3.4 Thermal spray powders. All powders shall be dry, free flowing and uniformly blended. Powders used for thermal spraying shall be designed specifically for thermal spraying, and the powder supplier shall specify and certify that:

a. All powdered materials have been inspected for particle size and certified that it is in accordance with the size range specified in the purchase order.

b. All powdered materials meet the chemical composition requirements as specified in the purchase order.

c. All powdered materials have been spray tested with intended application method. All coatings produced are smooth, uniform, free from cracks, voids, and have the required bond strength in accordance with customer requirements.

d. All powdered materials meet requirements of engineering and industry standards.

4.3.5 Gases. The following commercially acquired gases and gas specifications, or equivalent, shall be used for thermal spraying:

GAS	SPECIFICATION
Oxygen	CGA G-4
Acetylene	BB-A-106
Hydrogen	BB-H-886
Nitrogen	A-A-59155 or A-A-59503

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

4.3.5.1 Gases for plasma spraying. Gases for plasma spraying shall be as follows:

GAS	SPECIFICATIONS
Hydrogen	Pre-Purified 99.5%, Max-Oxygen Content 0.05% Max-Dewpoint -76 °F (-60 °C)
Nitrogen	Pre-Purified, Max-Oxygen Content 0.002% Max-Dewpoint -76 °F (-60 °C)
Argon	High Purity

5 DETAILED REQUIREMENTS

5.1 Thermal spray repair operations – Procedure. Thermal Spray coating and post treatment is usually the last repair operation made on a part, although thermal spray is also utilized in new manufacture. All cleaning, inspection, other than dimensional inspection, weld repairs, anodizing, peening, and electroplating repairs shall be performed and accepted prior to thermal spray coating, unless either a. or b. applies.

a. Thermal spray coating is suitably protected from subsequent repair operations.

b. Engineering Source Authority (ESA) has approved otherwise.

c. Areas to be sprayed shall have all previous coatings removed before spraying by either chemical or mechanical means. Unless otherwise specified, grit blasting shall not be an acceptable method of coating removal.

5.1.1 Part cleaning. Parts shall be cleaned following customer requirements in accordance with MIL-DTL-5002. Surface to be coated needs to be free of oils, inks, and other contaminants and debris.

5.1.2 Masking. Areas not to be grit blasted and coated, shall be masked. If selected areas are to be sprayed, then the surrounding areas shall be masked with approved tapes or masking materials. Operators shall not touch or contaminate surfaces to be coated with bare hands.

a. Table I contains the types of masking tapes that prevent part overspray during grit blasting and thermal spray coating. The masking tapes shall only be used when applicable for compatible thermal spray processes.

b. Sheet metal inserts or shielding, carbon plugs, Teflon™, silicone rubber, and other suitable materials are also permitted to be used for masking or locally fabricated masking templates, as long as it does not adversely interfere with the coating application/quality or damage the part.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

TABLE I. **Types of masking tapes.**

PMC 4235	Aluminum foil tape: is used for grit blasting and low heat areas.
PMC 4273	Fiberglass tape: is used to protect areas that are not near heat.
PMC 4295	Silicone tape: is used for high heat areas.
PMC 4399	Silicone-coated glass fabric: is used to protect parts during grit blasting and various thermal spray operations.
PMC 4404	Silicone-coated, glass fabric tape: is used to protect parts during grit blasting and various thermal spray operations.
PMC 4433	Aluminum foil, inert glass cloth tape with pressure-sensitive silicone adhesive: is used as a protective mask in high-temperature processes.
PMC 4458	Silicone-coated glass fabric: is used for protecting parts during grit blasting and various thermal spray operations.
PMC 4480	Silicone layer over aluminum: is used for high heat areas and excessive thermal spray coating build-ups.
PMC 4614	Glass cloth tape, coated with a pressure-sensitive silicone adhesive on both sides: is used as a protective mask in high-temperature processes.
PMC 4630	Silicone rubber/glass cloth tape with high temperature silicone adhesive: is used for protection from grit blast operations and thermal spray operations.
PMC 4639	Aluminum foil tape with pressure-sensitive silicone adhesive: is used as a protective mask with high-temperature processes.
PMC 4647	High-temperature, glass cloth, coated tape: is used to protect parts during grit blasting and thermal spray operations.
PMC 4664	Copper foil tape/glass cloth laminate: is used for high velocity oxygen fuel (HVOF) coating operations.
PMC 4667	High-temperature, glass cloth, coated tape: is used for high velocity oxygen fuel (HVOF) coating systems.

5.1.3 Abrasive grit blasting. Unless otherwise specified by the ESA, to remove oxidation and roughen the surface, the areas to be coated shall be grit blasted, and the manufacturer's recommendations followed. Surfaces shall be roughened to the extent that allows for the coating to achieve the bond strength required by the customer.

5.1.4 Preheating. Following customer and part requirements, surfaces to be coated shall be preheated to remove moisture and, when required, to control thermal expansion of the part with respect to coating. Preheat shall be accomplished by controlling torch dwell time immediately prior to spraying or by other suitable means. Temperature of part during preheat and subsequent spraying shall be maintained sufficiently low to prevent discoloration, oxidation, distortion, and other conditions detrimental to coating or base metal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

a. Special care shall be taken when coating parts to avoid overheating during spraying. Unless otherwise specified by the ESA, peak temperature of parts being sprayed shall be at least 50° F (10° C) below temper or aging temperatures of the substrates. Adjust as required for situations where the material plume crosses in-front of the temperature sensing site.

b. Gun-to-work (working) distance and spray angle shall remain controlled within designated parameters during surface preparation and coating.

5.1.5 Part cooling. Part and coating temperatures shall be controlled to prevent warpage and damage of the substrate and failure of the coating. Work stations shall have cooling systems allowing for proper temperature control, keeping the part under temper temperature and aging temperatures. Coating surface temperature shall be taken during and immediately after the part has been sprayed using a temperature gage, such as a laser sighted infrared thermometer or equivalent.

a. Unless otherwise specified by the ESA, the maximum temperature of the substrate during preheating and coating application shall be controlled not to exceed 350 °F (177 ºC) for most high strength steel surfaces, and 250 °F (121 °C) for aluminum substrates. Whenever possible, measure the surface being coated, or immediately adjacent to the surface being coated.

5.1.6 Spraying parameters. Unless otherwise specified by the ESA, for the spraying of each material, the air, fuel gas, primary and secondary pressures, flow settings, and working distance shall be in accordance with the manufacturer's recommendations.

5.1.7 Spraying. Coating material shall be deposited onto the designated surfaces to the thickness specified, or to a sufficient thickness to permit finishing to the dimensions specified. The manufacturer's recommendations for working distance shall be used for the material being sprayed. Bond coats shall be acceptable. Follow material manufacturer recommendations on application techniques and procedures.

a. The direction of the spray (spray angle) shall be as close to 90 degrees (perpendicular) with the substrate, as possible, and never less than 45 degrees. The coating shall be applied using multiple passes. Complete area coverage shall be made with each pass before the next pass is started. Coating thickness shall be continuously monitored while spraying to determine when the desired thickness has been achieved. Spraying shall continue until the coating thickness plus finishing allowance has been completed.

b. When repair illustration or instructions specify application of a bond coat, such coating(s) shall be applied as soon as practicable, preferably within two (2) hours after completion of previous coating, and shall be applied without intermediate surface preparation.

c. During application of plasma coating, equipment settings shall be continually monitored and adjusted to compensate for drift induced primarily by erosion of gun electrode and nozzle. Manufacturing tolerances make it extremely unlikely that any two (2) guns from any one (1) equipment manufacturer operate exactly alike. Atmospheric conditions shall be considered when they have subtle effects on plasma gun operation.

5.1.8 Visual inspection. Coatings shall fully bond with base metal. Coatings shall be uniform, continuous and free of spalling, chipping, flaking, cracks and other imperfections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

5.1.9 Removal of masking – Cleaning. After coating has been applied and part has cooled, masking materials shall be removed and residual masking adhesives cleaned off. Standard shop procedures and solvents shall be used. Care shall be taken to remove masking materials to not remove or damage the coating. Operators shall not touch or contaminate coating surfaces with bare hands.

5.1.10 Sealing sprayed coatings. Follow manufacturer's recommendations on when to use sealants, in conjunction with customer requirements.

5.1.11 Finish machining and grinding. After allowing the part to cool, overspray and material build-up shall be removed to prevent cracks from forming and extending into the main coating. Finish machining or grinding shall be performed, in accordance with MIL-STD-866 and manufacturer's recommendations, to the dimensions specified or as shown on the applicable reference documents.

5.1.12 Stripping. Follow the ESA's specific guidance. However, mechanical stripping by machining and grinding is generally approved. Chemical stripping may be used when applicable depending on substrate or coating system, in accordance with customer approvals.

5.2 Quality control.

5.2.1 Record Keeping. Records shall be kept in such a way that the process is repeatable regardless of operator. An example process operations sheet can be found in Appendix A *Sample plasma coating process operation sheet.* Automated records keeping is recommended.

5.2.2 Coating thickness. Coating thicknesses are specified by the applicable documents. If the thickness is not specified then the finish machine coating shall not exceed 0.030 inch (0.076 cm).

a. For grinding and machining, a minimum of 0.010 inch (0.025 cm) over the final dimension is typically accepted. Unless otherwise specified by the ESA, manufacturer's recommendations shall be followed.

b. For corrosion protection, minimum finished coating thickness shall be 0.003 inch (0.008 cm). Consult the ESA for exceptions.

5.2.3 Workmanship. The sprayed coating shall present a uniform appearance with a fine-to-medium granular finish. Surface defects on the coating shall be limited to nodules, and shall not exceed 0.045 inch (0.114 cm) in diameter or 0.025 inch (0.064 cm) above the surrounding sprayed surface. The following scenarios shall be cause for rejection:

a. Blisters, cracks, chips, or loosely adhering particles.

b. Pits that expose the undercoating or base metal.

c. Chips, cracks, or tears of the coating after final machining.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

5.2.4 Testing. The following tests shall be used as a method of controlling the quality of thermal spray coatings. Consult the ESA for more detailed descriptions of these tests, as needed, as well as for special cases and how to analyze results for each individual approved material for acceptance or re-test.

5.2.4.1 Bend test. Test specimens are made of rectangular panels with the following approximate dimensions: 1.00 x 3.00 x 0.05 inches (2.54 x 7.62 x 0.17 cm). Specimens shall be manufactured from identical material and heat-treat condition as components to be coated. Alternatively, specimen material shall be AMS 6454 or AMS 6351, heat-treated to produce a Rockwell C hardness greater than 40. Unless otherwise specified, specimens shall be coated on one (1) side to 0.003 - 0.005 inches (0.008 – 0.013 cm) thick. To reduce edge effects, the specimen shall have the coating deposited in such a manner as to leave a 0.5 inch (1.3 cm) wide coating strip. Specimens shall be tested by being bent around a 0.5 inch (1.3 cm) diameter bar or equivalent, with the coated surface on the outside of the bend. Specimens shall be bent to obtain at least a 90 degrees permanent set. See Figure 1 *Bend Test Specimen.*

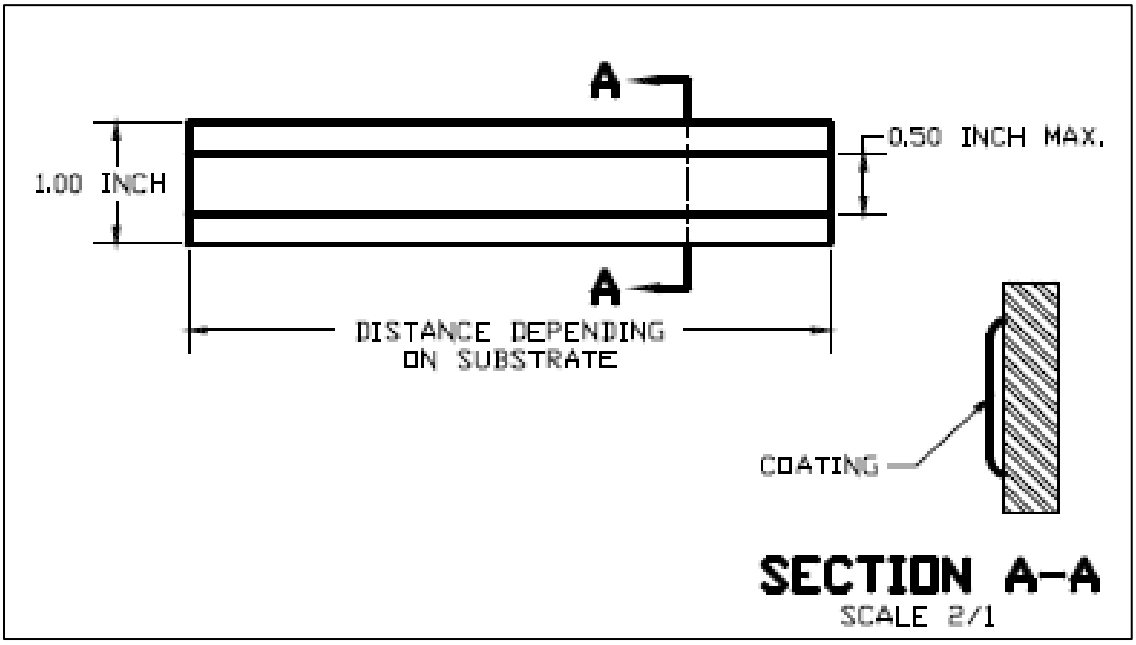


FIGURE 1. Bend test specimen.

5.2.4.1.1 Bend testing shall show no evidence of peeling, spalling, or separation of the coating from the substrate, when examined visually without magnification. Cracking of the coating shall be acceptable as depicted by Figure 2 *Acceptance standard for bend test specimens.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

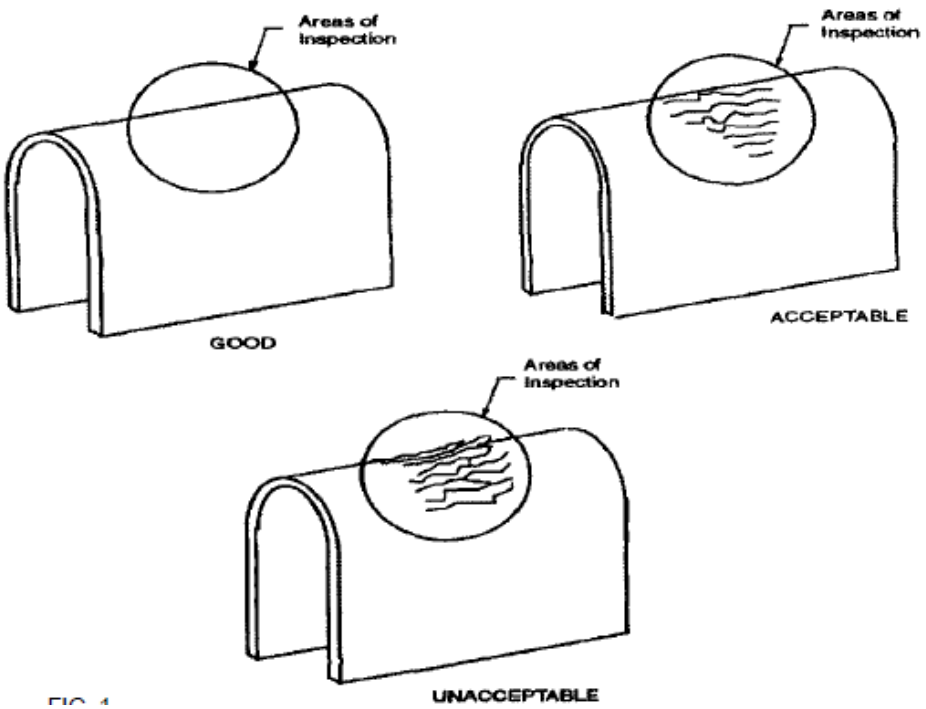


FIGURE 2. Acceptance standard for bend test specimens.

5.2.4.2 Bond strength. Test specimens, approximately 1.0 inch (2.5 cm) diameter by 1.5 inches (3.8 cm) minimum length shall be manufactured from identical material and heat-treat condition as components to be coated. Length may be reduced to 1.0 inch (2.5 cm), when required, to allow specimen reuse providing all coating and adhesives are removed and surface prepared for application accordingly. Alternatively, specimen material shall be as specified in ASTM C 633 or manufactured from SAE AMS 6484, heat-treated to produce a Rockwell C hardness greater than 40. Specimens shall be coated to a thickness of 0.009 inch (0.023 cm) to 0.012 inch (0.030 cm), prepared and tested in accordance with ASTM C 633. Adhesive only qualification tests shall be run under the same test conditions to verify integrity of the adhesive. FM© 1000 film adhesive is suitable for rough or roughed coatings up to 10 ksi (6.9 x 107 Pa). Other adhesives shall be used, as required, according to needs.

5.2.4.2.1 The ESA will specify bond strength values. If none is specified, bond strength shall be acceptable if finishing methods do not result in coating delamination.

5.2.4.3 Metallography. A detailed standard procedure for specimen preparation shall be used to ensure consistent results. Such as ASTM E 1920, but other standard procedures shall be used as needed, as well. A bend test specimen shall be used, as required, for metallographic evaluation. Examination of the entire cross-sectioned specimen at 200X (or greater) shall ensure that sample uniformity is consistent and show the coating to be free from cracks and delamination. Re-polishing shall only be performed on specimens that show flaws induced by cutting or polishing methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

5.2.4.3.1 Voids, oxides, and un-melted particles. Follow Manufacturer recommendations and ESA requirements.

5.2.4.3.2 Interface. Contamination of the coating or substrate interface with surface preparation media shall not exceed 10% at each field of view. Any coating separation at the interface shall not be acceptable. Separation is defined as a gap between the coating and substrate greater than 0.002 inch (0.005 cm) in length following directly along the bond line. Examine ten (10) fields of view at 200X magnification.

5.2.4.3.3 Porosity. The ESA will determine porosity.

5.2.4.3.4 Micro hardness. Test specimens, 1.00 x 3.00 x 0.05 inches (2.54 x 7.62 x 0.17 cm), shall be coated on one (1) side to a minimum thickness of 0.009 inch (0.023 cm). The hardness shall be the average of a minimum of ten (10) evenly spaced indentations in accordance with ASTM E384. Bend test or alternate design specimens such as small disks shall be acceptable.

5.2.4.3.5 Coating hardness. The ESA will determine coating hardness.

5.3 Quality. The coating of the finished component, as received by purchaser, shall be adherent to the basis material and shall have a uniform, continuous surface free from spalling, chipping, flaking, and other imperfections detrimental to usage of the coating.

5.3.1 Pre-production tests. All requirements of section 5.2.3 are acceptance tests and shall be performed prior to, or on the initial shipment of, coated parts.

5.3.2 Acceptance test frequency. The ESA will determine acceptance test frequency. If none are specified, test frequency shall be accomplished prior to beginning a new coating and substrate system, material lot changes, and any equipment alterations.

5.3.3 Acceptance test completion. Production spraying shall not be performed until the successful completion of all acceptance tests. Failure of any acceptance tests shall require a thorough evaluation and identification of the failure mechanism with implementation of corrective actions followed by a successful completion of all acceptance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

6 NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

6.1 Intended use. This document provides guidance for thermal spraying on aircraft components.

6.2 Subject term (key word) listing.

Flame spray
Gases
Procedures
Spray system
Testing
Types

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

APPENDIX A.

SAMPLE PLASMA COATING PROCESS OPERATION SHEET.

A.1 SCOPE

A1.1 Scope. The appendix provides an example of the information used as guidance when setting up equipment (4.2.13.c.). This appendix is not a mandatory part of this standard. The information contained herein is not intended for compliance.

PLASMA COATING PROCESS OPERATION SHEET

APPLICATION OF_____ **ON**_____

USING CONTROL PANEL_____ **GUN**_____

COATING MATERIAL

Powder_____

TEST PIECE MATERIAL

Steel, Nickel-based Alloys_____
Titanium, Titanium Alloys_____
Aluminum or Magnesium Alloys_____

SURFACE PREPARATION

Type of Grit_____
Grit Size_____
Air Pressure_____
Nozzle-to-Work Distance_____
Nozzle Size_____
Gun-to-Work Angle_____

SPRAY EQUIPMENT

Nozzle (Anode)_____
Cathode_____
Type of Gas (Primary)_____
Type of Gas (Secondary)_____
Type of Gas (Cover)_____
Cover of Gas Nozzle Number_____
Focal Point Relative to:
Nozzle Face_____

ARC GAS SETTINGS

Regulator PSI (Primary)_____ ± _____
Regulator PSI (Secondary)_____ ± _____
Back Pressure Gauge:
Operating PSI_____ ± _____
Cover Gas PSI_____ ± _____

PANEL FLOW

Gas (1)_____ ± _____
Gas (2)_____ ± _____
Cover Gas_____ ± _____

POWER

Voltage tap: Series_____
Parallel_____
Voltage DC Open Circuit_____
Voltage DC Operating_____ ± _____
Amperes DC Operating_____ ± _____
Power Control Setting: Start_____ ± _____
Run_____ ± _____

POWDER FEEDER DATA

Type_____
Carrier Gas_____
PSI_____ ± _____
Panel Flow_____ ± _____ CFH
Drive Gears_____
Dial Setting_____ ± _____
Meter Wheels_____
RPM_____ ± _____
Air Vibrator_____
PSI_____ ± _____
Feed Worm Pitch_____
RPM_____ ± _____
Vibrator Settings_____ ± _____
Powder Feed Rate_____
Grams/Min ± _____
Feeder Hose to Gun: Diameter ID_____
Length_____

COATING DATA

Powder Injection Port: Front_____
Rear_____
Gun-to-Work Distance_____ ± _____
Minimum Gun-to-Work Angle_____
Preheat Temperature_____

METHOD OF COOLING SUBSTRATE

Air Cool_____
Cooler Assembly_____
Air Pressure_____
Cross Jets_____

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH
Air Force – 70

Preparing Activity
Air Force – 70

(Project MFFP-2021-001)

Review Activities:
Navy – AS
Air Force – 20

NOTE: The activities listed above were interested in this standard as of the date of this standard. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T22:59Z
Check the source to verify that this is the current version before use.